Question 1: Prove the property from the lecture notes: given a continuous nonnegative random variable X with CDF F_X, show that

$$E[X] = \int_0^\infty (1 - F_X(t)dt$$

Equivalently, if X is a discrete nonnegative random variable, show that

$$E[X] = \sum_{k=0}^{\infty} P[X > k]$$

Question 2: The time X between customer arrivals at Tim Hortons has an exponential pdf with parameter λ, meaning that the the density of X is $f_X(t) = \lambda e^{-\lambda t}, x \geq 0$. Find the mean interarrival time.

Question 3: Given a random variable X with pdf $f_X(t) = \frac{1}{2}\lambda e^{-\lambda|t-\gamma|}$, compute its expectation.

Question 4: Show that $\mathrm{var[aX + b]} = \mathrm{a^2 var[X]}$ for any random variable and constants a, b. We recall that the variance is defined by $\mathrm{var[X]} = \mathrm{E[(X - E[X])^2]}$.

Question 5: For $\Omega = \{0, 1, 2, \ldots\}$ and X Geometric(p) where $p_X(k) = p(1-p)^k, k = 0, 1, 2, \ldots$, show that $E[X] = \frac{1-p}{p}$.

Question 6: A Mars rover fetching samples collects a total of 560 grams of material from four distinct sites and returns them to Earth in 4 canisters labeled A, B, C, D with 140, 112, 125, and 183 grams respectively, for further analysis. Note that, due to the sampling mechanism, the sample mass is always a discrete value in grams. First, a canister is randomly selected to make sure there is no damage on the canister. Let x denote the amount of material in the canister. Second, of all the returned material, a 1 gram sample is randomly extracted from the total material returned (not necessarily from the canister that was inspected). Let y denote the amount of material in the canister the sample was extracted from.

(I) Which of $E[x]$ and $E[y]$ do you think is larger? Why?

(II) Compute $E[x]$ and $E[y]$.

Question 7: Let N be a geometric random variable.

(I) Find $P[N > k]$.

(II) Find the cdf of N.

(III) Find $P[N$ is an even number].

(IV) Find $P[N = k|N \leq m]$.

Question 8: Suppose that the distribution of the diastolic blood pressure is normal for a randomly selected person in a certain population with mean 80 mm Hg and standard deviation 7mm Hg. If people with diastolic blood pressure 95 or above are considered hypertensive and people with diastolic blood pressure above 89 and below 95 are considered to have mild hypertension, what percent of that population have mild hypertension and what percent are hypertensive? Assume in that population no one has abnormal systolic blood pressure.

Question 9: Find the variance and standard deviation of a random variable X with distribution function:

$$F(x) = \begin{cases} 0 & x < -3 \\ 3/8 & -3 \leq x < 0 \\ 3/4 & 0 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

Is X a discrete or continous RV?

Question 10: For all α and $s > 0$, prove the following inequality:

$$P[X \geq \frac{\alpha}{s}] \leq e^{-\alpha}\mathcal{F}_X(-s)$$

where $\mathcal{F}_X(s) = \int_{-\infty}^{+\infty} e^{-sx} f_X(x)dx$ denores the Laplace transform of the pdf of X. [Hint: show that Markov's inequality for $Y = e^s X$ is applicable and these inequalities are equalent].

Question 11: If $E(\mathbf{x}) = 5$, and Var$(\mathbf{x}) = 5$, and $\mathbf{y} = a\mathbf{x} + b$, then find a and b such that $P\left[|\mathbf{y} - 10| \geq 0.1\right] \leq 0.01$.

Hint: Use Chebyshev inequality.